

Questions with Answer Keys

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Q1 - Single Correct

If in $\triangle ABC$, $b = c$ and $\angle A = 20^\circ$, then the value of $a^3 + b^3$ is

- (1) $3a^2b$
- (2) $3b^2c$
- (3) $3c^2a$
- (4) abc

Q2 - Single Correct

In $\triangle ABC$, $a = 3$, $b = 5$ and a, b, c are in AP , then the maximum value of $\tan A \tan B$ is

- (1) $\frac{1}{2}$
- (2) $\frac{1}{3}$
- (3) $\frac{1}{4}$
- (4) $\frac{1}{5}$

Q3 - Single Correct

In $\triangle ABC$, $\angle A = 60^\circ$, $\angle B = 45^\circ$ and let D divides AB internally in $1 : 3$. Then, $\frac{\sqrt{6} \sin(\angle ACD)}{\sin(\angle BCD)}$ is equal to

- (1) $\frac{1}{\sqrt{6}}$
- (2) $\sqrt{6}$
- (3) 1
- (4) None of these

Q4 - Single Correct

If two adjacent sides of a cyclic quadrilateral are 2 and 5 and the angle between them is 60° . If the area of the quadrilateral is $4\sqrt{3}$ sq units, then the remaining two sides are

- (1) 2, 3
- (2) 1, 2

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(3) 3,4

(4) 2,2

Q5 - Single Correct

If a, b, c, d are sides of a quadrilateral, then the value of $\frac{a^2+b^2+c^2}{d^2}$ is greater than

(1) 1

(2) $\frac{1}{2}$

(3) $\frac{1}{3}$

(4) $\frac{1}{4}$

Q6 - Single Correct

If p is the product of sines of angles of a triangle and q be the product of their cosines, then the tangents of the angle are roots of the equation

(1) $qx^3 - px^2 + (1+q)x - p = 0$

(2) $px^3 - qx^2 + (1+p)x - q = 0$

(3) $(1+q)x^3 - px^2 + qx - q = 0$

(4) None of the above

Q7 - Single Correct

Let $F(x+y) = F(x) \cdot F(y), \forall x, y$ and $F(1) = 2$. If in a $\triangle ABC, a = F(3), b = F(1) + F(3)$ and $c = F(2) + F(3)$, then $2A$ is

(1) C

(2) $2C$

(3) $3C$

(4) $4C$

Q8 - Single Correct

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In $\triangle ABC$, if r_1, r_2, r_3 be the exradii and $\frac{bc}{r_1} + \frac{ca}{r_2} + \frac{ab}{r_3} = \lambda \left[s \sum \frac{1}{a} - 3 \right]$, then λ is equal to

- (1) $\frac{abc}{\Delta}$
- (2) $\frac{2abc}{\Delta}$
- (3) abc
- (4) None of these

Q9 - Single Correct

In $\triangle ABC$, if $\cos A + \cos B + \cos C = \frac{7}{4}$, then $\frac{27R}{r}$ is equal to

- (1) 26
- (2) 36
- (3) 18
- (4) None of these

Q10 - Single Correct

The cosine of the obtuse angle formed by medians drawn from the vertices of acute angles of an isosceles right angled triangle is

- (1) $\frac{4}{5}$
- (2) $\frac{3}{5}$
- (3) $\frac{-4}{5}$
- (4) None of these

Q11 - Single Correct

A triangle is inscribed in a circle. The vertices of the triangle divide the circle into three arcs of lengths 3, 4 and 5 units, then the area of triangle is

- (1) $\frac{9\sqrt{3}(1+\sqrt{3})}{\pi^2}$ sq units
- (2) $\frac{8\sqrt{3}(1+\sqrt{3})}{\pi^2}$ sq units
- (3) $\frac{9\sqrt{3}(1+\sqrt{3})}{\pi}$ sq units

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(4) None of these

Q12 - Multiple Correct

In ΔABC , if $\frac{a \cos A + b \cos B + c \cos C}{a \sin B + b \sin C + c \sin A} = \frac{a+b+c}{9R}$, then

(1) $\angle A = 60^\circ$

(2) $\angle B = 60^\circ$

(3) $\angle C = 60^\circ$

(4) None of these

Q13 - Multiple Correct

Let ABC be an isosceles triangle with base BC . If r is the radius of the circle inscribed in ΔABC and r_1 is the radius of the circle escribed opposite to the $\angle A$, then the product $r_1 \cdot r$ can be equal to

(1) $R^2 \sin^2 A$

(2) $R^2 \sin^2 2B$

(3) $\frac{1}{2}a^2$

(4) $\frac{a^2}{4}$

Q14 - Multiple Correct

Lengths of the tangent from A, B and C to the incircle are in AP, then

(1) r_1, r_2, r_3 are in HP

(2) r_1, r_2, r_3 are in AP

(3) a, b, c are in AP

(4) $\cos A = \frac{4c-3b}{2c}$

Q15 - Multiple Correct

If p is the length of median from the vertex A to side BC of a ΔABC , then

(1) $4p^2 = 2b^2 + 2c^2 - a^2$

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$$(2) 4p^2 = b^2 + c^2 + 2bc \cos A$$

$$(3) 4p^2 = a^2 + 4bc \cos A$$

$$(4) 4p^2 = (2s - a)^2 - 4bc \sin^2 \frac{A}{2}$$

Q16 - Multiple Correct

If a, b, A be given in a $\triangle ABC$ and c_1 and c_2 be two possible values of third side such that $c_1^2 + C_1 C_2 + c_2^2 = a^2$, then $\angle A$ is

$$(1) 30^\circ$$

$$(2) 60^\circ$$

$$(3) 90^\circ$$

$$(4) 120^\circ$$

Q17 - Multiple Correct

In a $\triangle ABC$, if the radii of ex-circles r_1, r_2, r_3 are given by $r_1 = 8, r_2 = 12, r_3 = 24$, then

$$(1) r = 8$$

$$(2) a = 12$$

$$(3) b = 16$$

$$(4) c = 20$$

Q18 - Multiple Correct

If A, A_1, A_2, A_3 be the areas of the inscribed and escribed of a $\triangle ABC$, then

$$(1) \sqrt{A_1} + \sqrt{A_2} + \sqrt{A_3} = \sqrt{\pi} (r_1 + r_2 + r_3)$$

$$(2) \frac{1}{\sqrt{A_1}} + \frac{1}{\sqrt{A_2}} + \frac{1}{\sqrt{A_3}} = \frac{1}{\sqrt{A}}$$

$$(3) \frac{1}{\sqrt{A_1}} + \frac{1}{\sqrt{A_2}} + \frac{1}{\sqrt{A_3}} = \frac{s^2}{\sqrt{\pi r_1 r_2 r_3}}$$

$$(4) \sqrt{A_1} + \sqrt{A_2} + \sqrt{A_3} = \sqrt{\pi} (4R + r)$$

Q19 - Multiple Correct

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If in $\triangle ABC$, $r_1 = 2r_2 = 3r_3$, then

(1) $\frac{a}{b} = \frac{4}{5}$

(2) $\frac{a}{b} = \frac{5}{4}$

(3) $\frac{a}{c} = \frac{3}{5}$

(4) $\frac{a}{c} = \frac{5}{3}$

Q20 - Multiple Correct

If in $\triangle ABC$, a, b, c are in AP and p_1, p_2, p_3 are the altitudes from the vertices A, B, C respectively, then

(1) p_1, p_2, p_3 are in AP

(2) p_1, p_2, p_3 are in HP

(3) $p_1 + p_2 + p_3 \leq \frac{3R}{\Delta}$

(4) $\frac{1}{p_1} + \frac{1}{p_2} + \frac{1}{p_3} \leq \frac{3R}{\Delta}$

Q21 - Multiple Correct

Given an isosceles triangle with equal sides of length b , base angle $\alpha < \frac{\pi}{4}$ and R, r the radii and O, I the centres of the circumcircle and incircle respectively. Then,

(1) $R = \frac{1}{2}b \operatorname{cosec} \alpha$

(2) $\Delta = 2b^2 \sin 2\alpha$

(3) $r = \frac{b \sin 2\alpha}{2(1 + \cos \alpha)}$

(4) $OI = \left| \frac{b \cos \frac{3\alpha}{2}}{2 \sin \alpha \cos \frac{\alpha}{2}} \right|$

Q22 - Multiple Correct

Which of the following is/are not true in a $\triangle ABC$?

(1) $(b + c) \sin \left(\frac{B-C}{2} \right) = 2a \cos \frac{A}{2}$

(2) $(b + c) \cos \frac{A}{2} = 2a \sin \left(\frac{B-C}{2} \right)$

(3) $(b - c) \cos \frac{A}{2} = a \sin \left(\frac{B-C}{2} \right)$

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$$(4) (b - c) \sin\left(\frac{B-C}{2}\right) = 2a \cos \frac{A}{2}$$

Q23 - Multiple Correct

If a, b , and c are the sides of a $\triangle ABC$ such that $\frac{a+b}{11} = \frac{b+c}{13} = \frac{c+a}{12}$ and $\frac{\cos A}{p} = \frac{\cos B}{q} = \frac{\cos C}{r}$, then

$$(1) p = 25$$

$$(2) q = 19$$

$$(3) r = 7$$

(4) None of these

Q24 - Paragraph 1

Passage I (For Question 24, 25)

A polygon has n -sides. If all sides and all angles are same, then the polygon is called regular polygon. Let

$A_1, A_2, A_3, \dots, A_n$ are the vertices of a regular polygon of n -sides. R be the radius of circumscribed circle of regular polygon and r be the radius of inscribed circle of regular polygon and

$$A_1A_2 = A_2A_3 = A_3A_4 = \dots = A_nA_1 = a.$$

On the basis of above information, answer the following questions:

The value $r + R$ is

$$(1) \frac{a}{2} \cot\left(\frac{\pi}{n}\right)$$

$$(2) a \cot\left(\frac{\pi}{2n}\right)$$

$$(3) \frac{a}{4} \cot\left(\frac{\pi}{2n}\right)$$

$$(4) \frac{a}{2} \cot\left(\frac{\pi}{2n}\right)$$

Q25 - Paragraph 1

The area of a regular polygon of n -sides is

$$(1) \frac{1}{2} n R^2 \sin\left(\frac{2\pi}{n}\right)$$

$$(2) n R^2 \tan\left(\frac{\pi}{n}\right)$$

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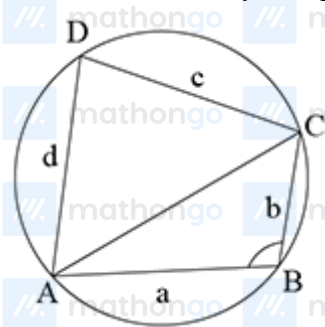
(3) $\frac{nr^2}{2} \sin\left(\frac{2\pi}{n}\right)$

(4) $nr^2 \tan\left(\frac{2\pi}{n}\right)$

Q26 - Paragraph 2

Passage II (For Question 26, 27)

Let $ABCD$ be a cyclic quadrilateral inscribed in a circle of radius R .



Then, $\cos B = \frac{a^2 + b^2 - c^2 - d^2}{2ab + 2cd}$

and area of quadrilateral $= \frac{1}{2}(ab + cd) \sin B$

$= \sqrt{(s-a)(s-b)(s-c)(s-d)}$

when, $s = \frac{a+b+c+d}{2}$

Also, $AC^2 \cdot BD^2 = (ac + bd)^2$

i.e. $AC \cdot BD = AB \cdot CD + BC \cdot AD$

and $R = \frac{AC}{2 \sin B}$

On the basis of above information, answer the following questions:

The sides of quadrilateral which can be inscribed in a circle are 6, 6, 8 and 8 cm. Then, radius of circumcircle is

(1) $\frac{5}{12}$ cm

(2) $\frac{24}{7}$ cm

(3) $\frac{11}{7}$ cm

(4) None of these

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Q27 - Paragraph 2

The sides of a quadrilateral which can be inscribed in a circle are 5, 5, 12 and 12 cm. Then, the radius of incircle of the quadrilateral is

- (1) $\frac{15}{17}$ cm
- (2) $\frac{30}{17}$ cm
- (3) $\frac{60}{17}$ cm
- (4) incircle does not exist

Q28 - Paragraph 3

Passage III (For Question 28, 29)

Consider a $\triangle ABC$, where x, y, z are the length of perpendicular drawn from the vertices of the triangle to the opposite side a, b, c respectively and let the letters R, r, s, Δ denote the circumradius, inradius, semi-perimeter and area of the triangle respectively.

On the basis of above information, answer the following questions:

If $\frac{bx}{c} + \frac{cy}{a} + \frac{az}{b} = \frac{a^2+b^2+c^2}{k}$, then the value of k is

- (1) R
- (2) s
- (3) $2R$
- (4) $\frac{3}{2}R$

Q29 - Paragraph 3

If $\cot A + \cot B + \cot C = k \left(\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} \right)$, then the value of k is

- (1) R^2
- (2) Rr
- (3) Δ
- (4) $a^2 + b^2 + c^2$

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Q30 - Paragraph 4

Passage IV (For Question 30, 31)

In $\triangle ABC$ with internal angles A, B and C , points O, I and H respectively denote circumcentre, incentre and orthocentre. Given that the points A, H, I, O and B are concyclic. On the basis of above information, answer the following questions:

Angle C is equal to

- (1) $\frac{\pi}{6}$
- (2) $\frac{\pi}{4}$
- (3) $\frac{\pi}{3}$
- (4) $\frac{5\pi}{12}$

Q31 - Paragraph 4

The ratio $HI : IO$ is equal to

- (1) $\frac{1}{2}$
- (2) 1
- (3) $\sqrt{2}$
- (4) $\sqrt{3}$

Q32 - Paragraph 5

Passage V (For Question 32, 33)

In a triangle, if sum of two sides is p and product is q ($p \geq 2\sqrt{q}$) such that $(p+r)(p-r) = q$, where r is the third side of a triangle.

On the basis of above information, answer the following questions:

The greatest angle of triangle is

- (1) 105°

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(2) 120° (3) 135° (4) 150°

Q33 - Paragraph 5

The area of triangle is

(1) $\frac{\sqrt{3}}{4}q$ sq units(2) $\frac{\sqrt{3}}{4}p$ sq units(3) $\frac{\sqrt{3}}{4}r$ sq units

(4) None of these

Q34 - Integer Type

The sides of a cyclic quadrilateral are in AP, the shortest side is 6 and the difference of the longest side and the shortest side is also 6. Let A be the area of quadrilateral such that $A^2 = 1152\lambda$, then λ is

Q35 - Integer Type

In a $\triangle ABC$, if $\cot \frac{A}{2} \cot \frac{B}{2} = c$, $\cot \frac{B}{2} \cot \frac{C}{2} = a$ and $\cot \frac{C}{2} \cot \frac{A}{2} = b$, then the value of $\frac{1}{s-a} + \frac{1}{s-b} + \frac{1}{s-c}$ is

Q36 - Integer Type

Let $A_1, A_2, A_3, \dots, A_n$ be the vertices of an n -sided regular polygon such that $\frac{1}{A_1A_2} = \frac{1}{A_1A_3} + \frac{1}{A_1A_4}$, then the value of n is

Q37 - Integer Type

Let ABC and ABC' be two non-congruent triangles with sides $AB = 4$, $AC = AC' = 2\sqrt{2}$ and $\angle B = 30^\circ$.

The absolute value of the difference between the areas of these triangles is

Q38 - Integer Type

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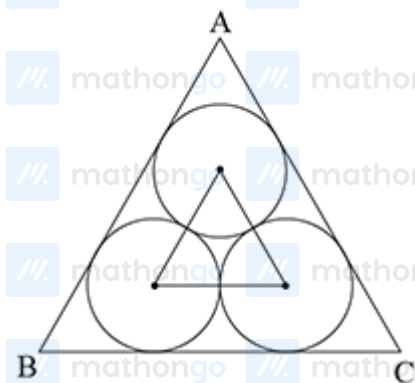
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If r and R be respectively the radii of inscribed and circumscribed circles of a regular polygon of n -sides such that $\frac{R}{r} = \sqrt{5} - 1$, then n is equal to

Q39 - Integer Type

If the area of equilateral $\triangle ABC$ which contains three coins of unit radius, is $6 + 4\sqrt{p}$. Then, the value of p is



Q40 - Integer Type

In $\triangle ABC$, if $\frac{r_1}{6} = \frac{r_2}{3} = \frac{r_3}{2}$ and $\frac{a}{b} + \frac{b}{c} + \frac{c}{a} = \frac{191}{20\lambda}$, then the value of λ is

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